

Samuele Marro

Head @ IDAI

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I'm a PhD student at Oxford under Philip Torr, and I run the Institute for Decentralized AI, a joint Oxford-Stanford research institute. The institute is funded by 1M USD in grants by OpenAI, Anthropic, Google, Microsoft, and the UK ARIA. I lead a team of 15 people to build the new generation of successfully decentralized AI systems.

Research Experience

- Aug '25 – Now **Head, Institute for Decentralized AI.**
- Head of a research institute focused on security and infrastructure for decentralized and distributed AI systems.
 - Two main units: Oxford and Stanford.
 - Funded by the **AI Safety Fund** and the **Cosmos Institute**.
- Apr '26 – Jul '26 **PhD Research Intern, Microsoft.**
- PhD internship in Microsoft's Agents Team of the Applied Sciences Group.
 - Main focus: developing efficient local agents.
- Nov '23 – Jul '24 **Predoctoral Research Fellow, University of Bologna.**
- Department fellowship on adversarial robustness under Prof. Michele Lombardi.
 - Proved that adversarial defenses for neural networks are Σ_2^P -hard (i.e., computationally infeasible).

Selected Publications

- Feb '26 **Samuele Marro**, Jialin Yu, Emanuele La Malfa, et al., *Benchmarking at the Edge of Comprehension*, **ICML 2026 (Oral)**.
- May '25 Emanuele La Malfa, Gabriele La Malfa, **Samuele Marro**, et al., *Large Language Models Miss the Multi-Agent Mark*, **NeurIPS 2025 (Oral)**.
- Jan '25 Tobin South, **Samuele Marro**, Thomas Hardjono, et al., *Authenticated Delegation and Authorized AI Agents*, **ICML 2025 (Oral)**.
- Sep '24 **Samuele Marro**, Davide Evangelista, Emanuele La Malfa, et al., *Language Models Are Implicitly Continuous*, **ICLR 2025**.
- Jul '23 **Samuele Marro**, Michele Lombardi, *Computational Asymmetries in Robust Classification*, **ICML 2023**.

Selected Awards

- Aug '26 **Grant, UK Advanced Research and Innovation Agency, 550k GBP.**
- Grant by the Scaling Trust division of the Advanced Research and Innovation Agency on automated algorithmic game design.
- Aug '25 **Grant, AI Safety Fund, 450k USD.**
- Grant by the AI Safety Fund (joint **Google-OpenAI-Anthropic-Microsoft** fund) on the decentralized oversight of agent networks.
- Mar '24 **Microsoft Studentship, 250k USD.**
- Studentship for research into communication protocols between LLMs and multimodal memory in generative models.

Education

- Oct '24 – Today **DPhil in Artificial Intelligence, University of Oxford.**
- 4-year DPhil at the Oxford AIMS CDT under Prof. **Philip Torr** on **multiagent LLMs**.
 - Funded by a 250k USD Microsoft studentship. Graduation: Oct '28 (can be anticipated).
- Sep '22 – Jul '24 **MSc in Artificial Intelligence, University of Bologna.**
- GPA: **29.75/30**, top of the class (170 students). Graduation grade: **110 cum laude**.
- Oct '19 – Jul '24 **Excellency Degree (COTS), Institute of Superior Studies.**
- 5-Year excellency degree (EQF8, Italian system). Equivalent to a Master's Degree.
 - GPA: **29.88/30**. Research topic: adversarial robustness.
- Oct '19 – Jul '22 **BSc in Computer Science, University of Bologna.**
- GPA: **29.77/30**, top of the class (150 students). Graduation grade: **110 cum laude**.

Other Awards

- Dec '25 **Exploration Grant**, *UK Advanced Research and Invention Agency*, 20k GBP.
 - Exploratory grant on automated protocol design as a path to full cybersecurity resilience.
 - Part of the “Trust Everything Everywhere” research area.
- Nov '23 **AiXIA “Leonardo Lesmo” Award**.
 - Award for the best MSc thesis in AI in Italy, granted by the Italian Association for Artificial Intelligence.
 - I accidentally submitted my BSc thesis, but won anyway.
- Mar '21, Mar '22 **Hackathon Prizes**, *Gitcoin Hackathons*, 7k USD.
 - 6k USD won by developing Solidity smart contracts for the Vite blockchain.
 - 1k USD won by developing tooling to improve IPFS usage.
- Jan '21 **CINECA HPC Grant**, 170k Core-Hours.
 - Grant for the Galileo100 HPC cluster for research on the computational complexity of adversarial attacks.

Other Publications

- Feb '26 **Samuele Marro**, Philip Torr, *LLM Agents Are the Antidote to Walled Gardens*, **ICML 2026**.
- Jan '26 **Samuele Marro**, Alan Chan, Xinxing Ren, et al., *Permission Manifests for Web Agents*.
- Nov '24 X Angelo Huang, Emanuele La Malfa, **Samuele Marro**, et al., *A notion of complexity for theory of mind via discrete world models*, **EMNLP 2024**.
- Oct '24 **Samuele Marro**, Emanuele La Malfa, Jesse Wright, et al., *A Scalable Communication Protocol for Networks of Large Language Models*.
- Dec '23 Andrea Asperti, Davide Evangelista, **Samuele Marro**, et al., *Image embedding for denoising generative models*, **Artificial Intelligence Review**.

Other Experience

- Sep '25 – Now **Fellow**, *Cosmos Institute*.
 - Research fellow on the intersection of AI, philosophy, and decentralization.
 - Goal: build stable non-governmental institutions to steer AI towards a less authoritarian future.
- Jan '22 – Dec '24 **Solidity Developer and Auditor**, *Freelance*.
 - Performed Smart Contract audits, Web3 integrations and secure Smart Contract development.
 - Specialized in EVM and EVM-like (e.g. Vite) security standards.
- Nov '21 – Mar '22 **Machine Learning Intern**, *MindIt Srl + Kantar*.
 - Developed a **Tensorflow** LSTM implementation compatible with **IBM CPLEX** optimization algorithms.
 - Rewrote Meta's FBProphet time series model in **Tensorflow**.
- Sep '22 – Jan '23 **Teaching Assistant**, *University of Bologna*.
 - Taught the Applied ML module of “Introduction to Machine Learning” for the Fall '22 semester.
- Jun '17 – Jul '17 **Software Development Intern**, *Sistemi Tre*, Alba, Italy.
 - Automated database queries through the in-house software interface using **AHK**, leading to a 3-4x increase in data retrieval speed.
 - Assisted IT staff during on-site technical support.

Affiliations

- Mar '25 – Today **Chair**, *Lightweight Agent Standards Working Group*.
 - Industry-academia joint working group, focused on designing and iterating quickly on standards for agent interactions.
 - Notable organizations: Oxford University, Stanford University, MIT, Firecrawl.
- Jan '26 – May '26 **Mentor**, *Cambridge ERA Fellowship*.
 - Mentored fellows on applications of multi-agent security techniques to AI safety.
- Mar '23 – Nov '25 **Head**, *Computer Science Student Commission, Superior College of Bologna (ISS)*.
 - Supervised CS student projects and advised students on CS and ML-related research directions.
- Jul '22 – Jul '25 **Representative**, *CS Department, University of Bologna*.
 - Managed student-department relations and represented student interests in department meetings.